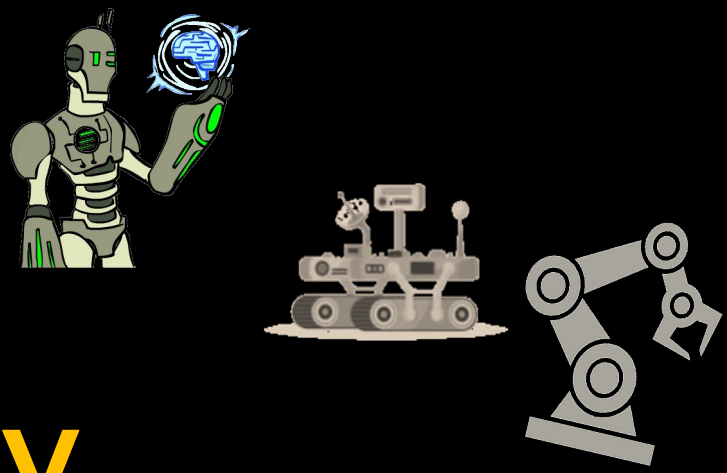
AI WEEK 2022

منظمة الذكاء الاصطناعي
Artificial Intelligence Organization



Safe and Trustworthy Artificial Intelligence

جامعة قطر
QATAR UNIVERSITY

JUNAID QADIR

Various use cases of AI predictions

REINFORCEMENT LEARNING





SELF-DRIVING CARS

IMAGE CLASSIFICATION





HEALTH INFORMATICS



OBJECT RECOGNITION



Machine translation: English sentence → Japanese sentence



Dialogue: conversational history → next utterance



Image segmentation: image → segmentation

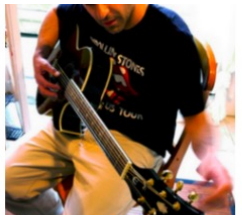
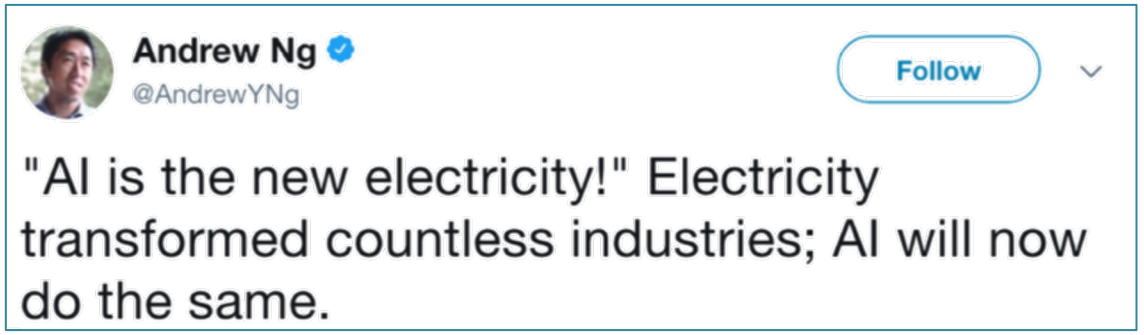


Image captioning: image → sentence describing image



Different takes on “what is intelligence”

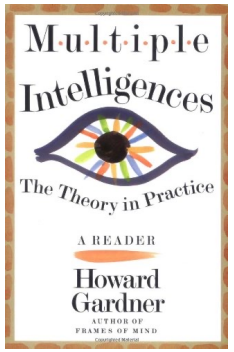
The ability to solve problems

The ability to learn from experience

The ability to reason about things

The ability to recognize patterns



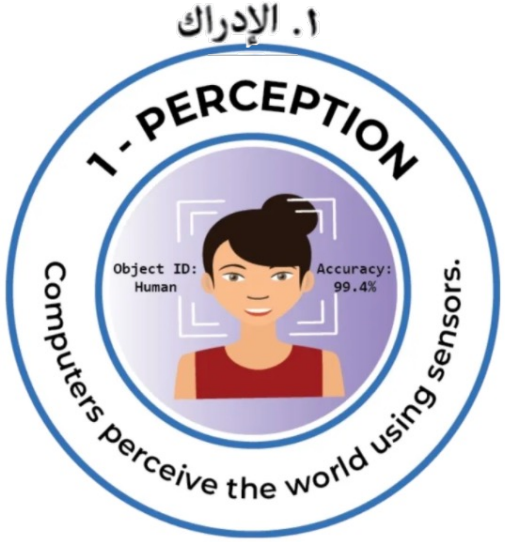


The ability to figure out causes of things (or to understand)

The ability to generalize and abstract (and make analogies)

The ability to pursue objectives and purpose.

Big Ideas in Artificial Intelligence



Big Idea 1 in AI



Big Idea 2 in AI



Big Idea 3 in AI



Big Idea 4 in AI



Big Idea 5 in AI

https://ai4k12.org/

The AI Dilemma

(Can we trust that AI applications will be used for human-beneficial ends and not to harm people)



Open questions related to AI

Bias (Can the AI model learn our biases?)

Interpretability (Can we understand AI models' "reasoning"?)

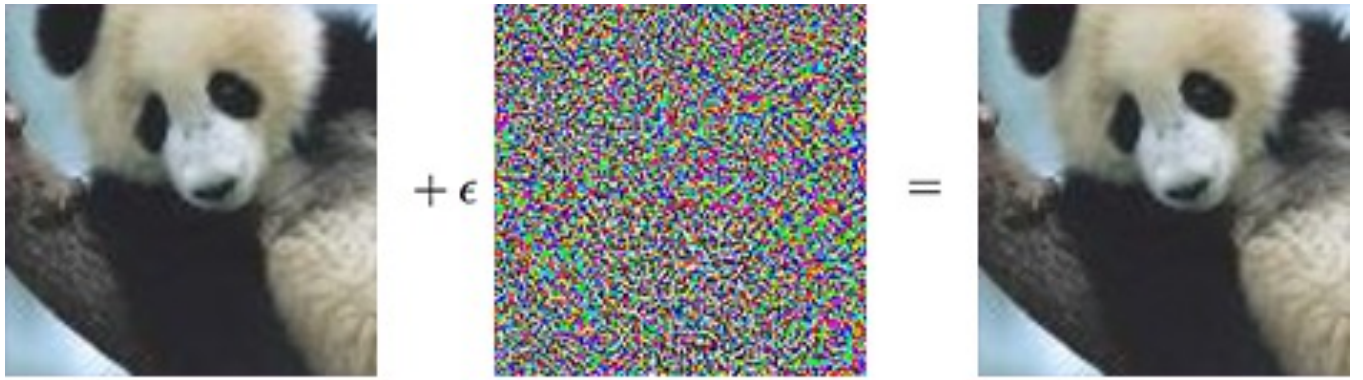
Privacy (Is the AI model protecting our privacy?)

Security (Is the AI model secure against adversarial attacks?)

Trustworthiness/Reliability

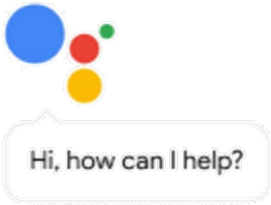
(Can I trust the prediction of an AI model?)

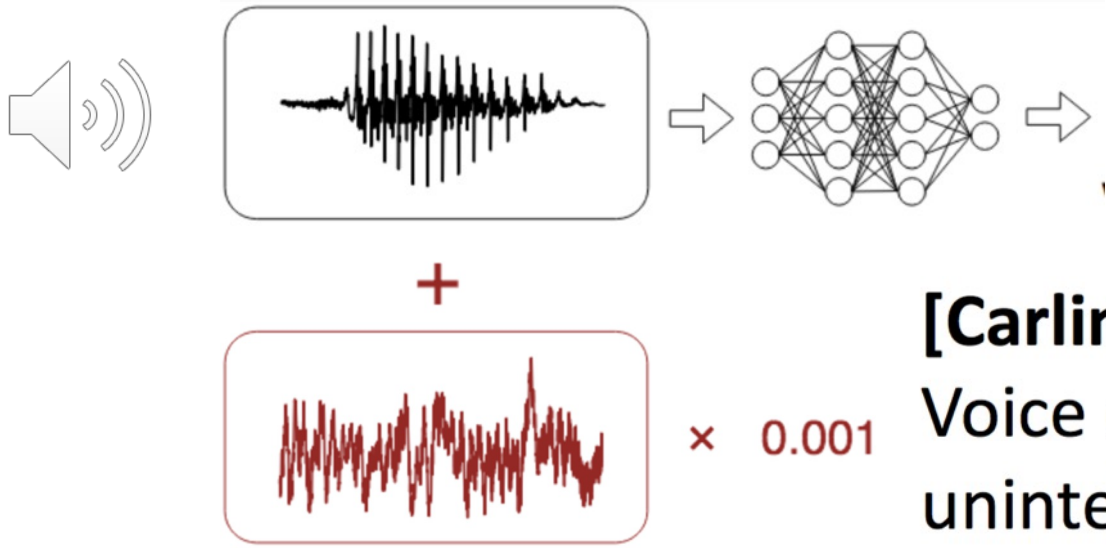
Adversarial attacks on AI models





*Adversarial examples are inputs to AI models that an **attacker has intentionally designed** to cause the model to make a mistake*





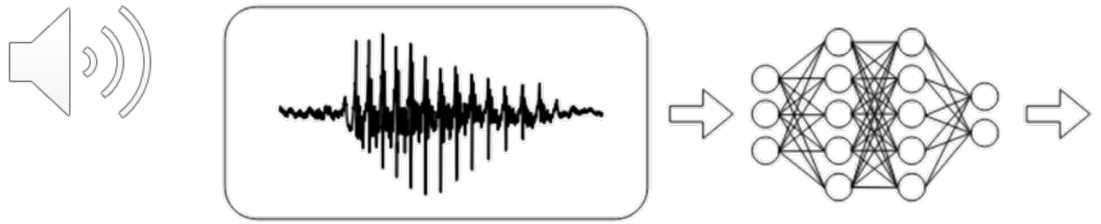
"without the dataset the article is useless"

+

× 0.001

[Carlini Wagner 2018]:
Voice commands that are unintelligible to humans

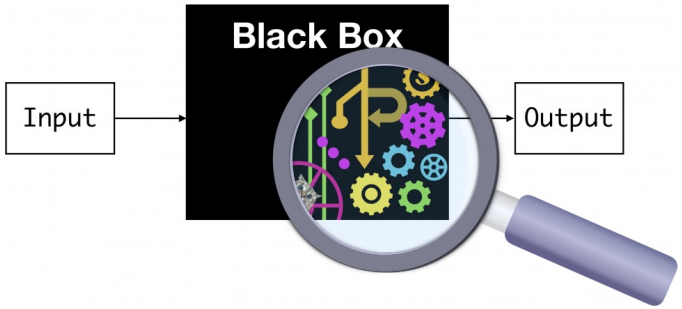
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"okay google browse to evil dot com"



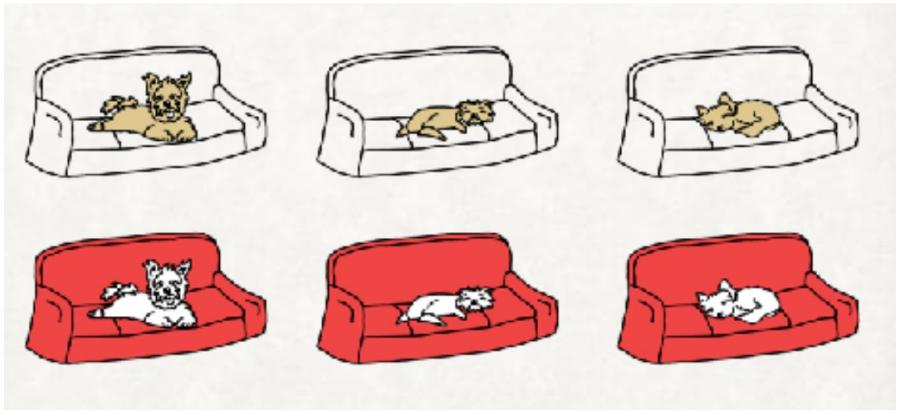
Why does AI fail in this way?





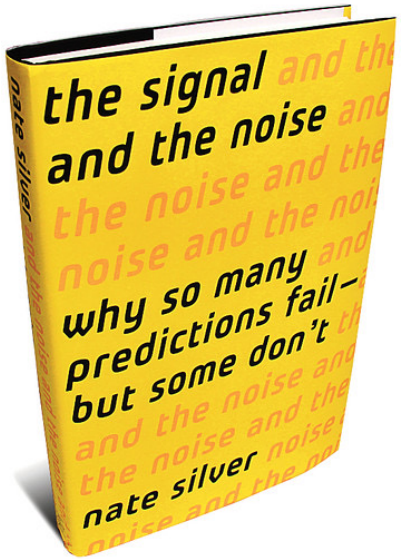
Wilhelm von Osten and Clever Hans. What is the horse looking at? Credit: Chronicle/Alamy Stock Photo

Clever Hans (pictured) "solved" arithmetic problems by simply following the cues that inadvertently emanated from his trainer's body language.

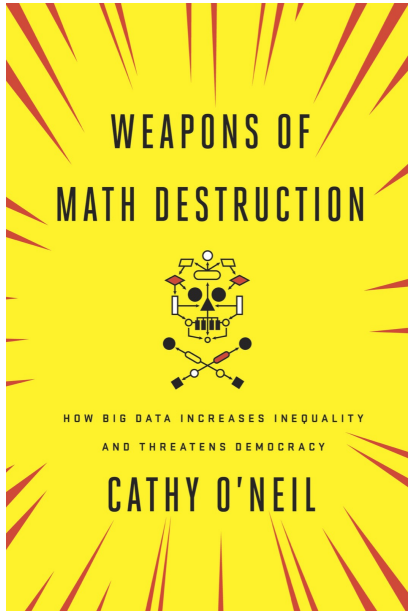


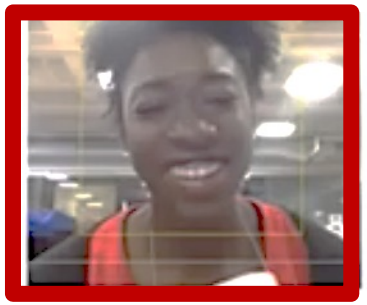


FAIL



The problem of bias: racial bias







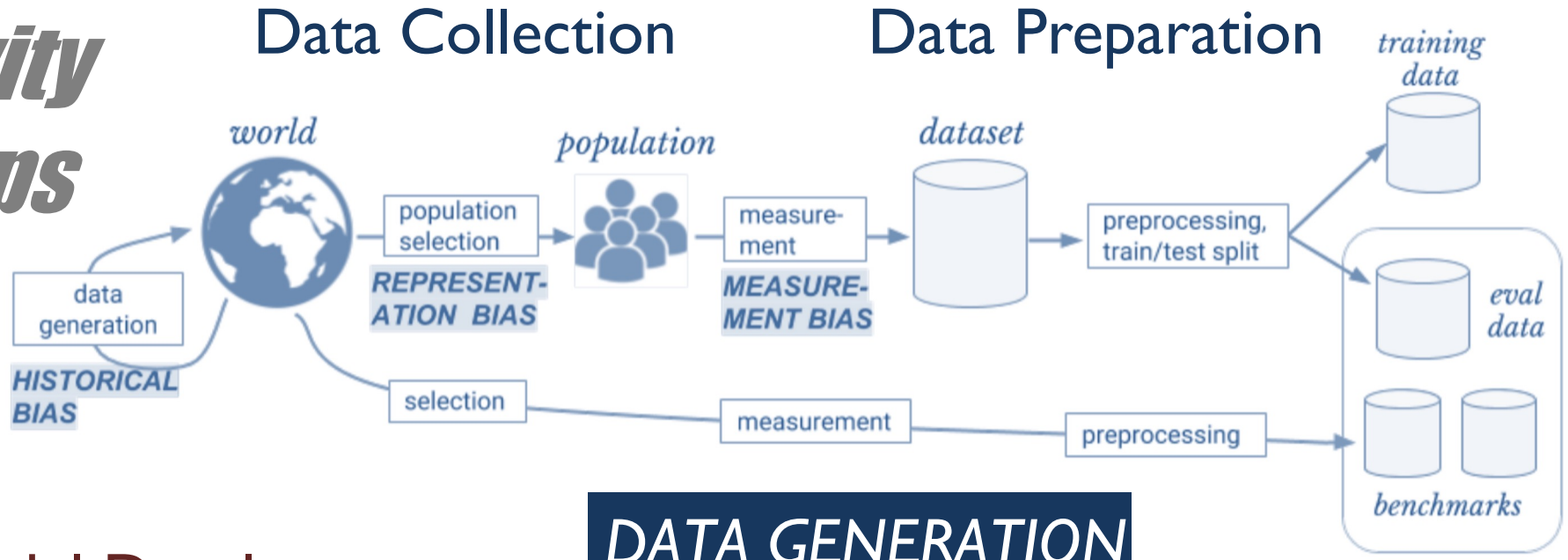




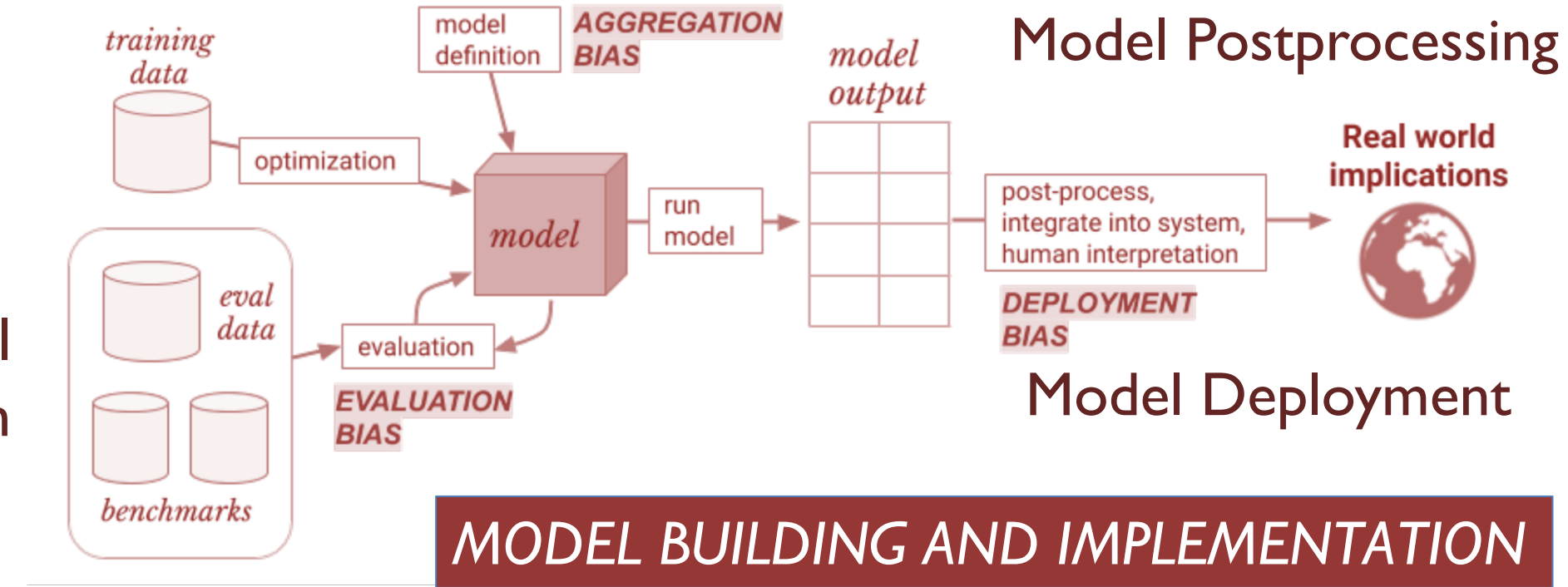
PROPUBLICA



How Subjectivity and Bias Creeps into AI

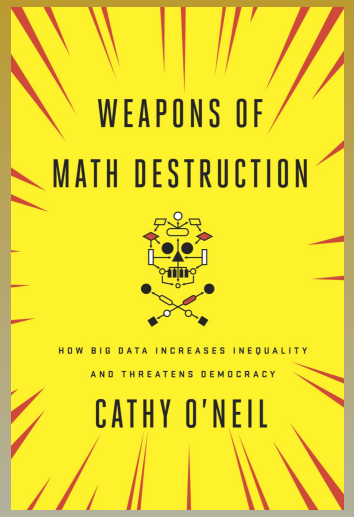


Suresh and Guttag, *A Framework for Understanding Unintended Consequences of Machine Learning* https://arxiv.org/pdf/1901.10002.pdf



“

THERE ARE ETHICAL CHOICES IN EVERY SINGLE ALGORITHM WE BUILD



How then to develop .. Safe & Trustworthy AI

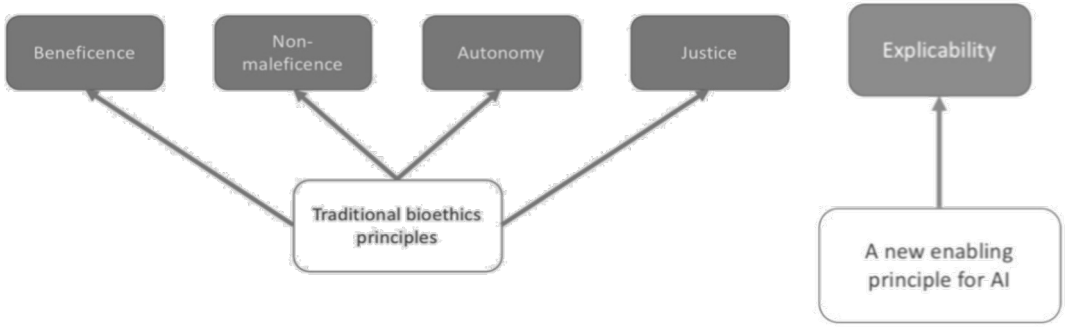
Prosocial AI requires transdisciplinary work

nature machine intelligence
PERSPECTIVE
https://doi.org/10.1038/s42256-019-0088-2

The global landscape of AI ethics guidelines

Anna Jobin, Marcello Ienca and Effy Vayena*

The four high-level ethical principles capture the essence of most AI declarations



nature machine intelligence
PERSPECTIVE
https://doi.org/10.1038/s42256-019-0114-4

Principles alone cannot guarantee ethical AI

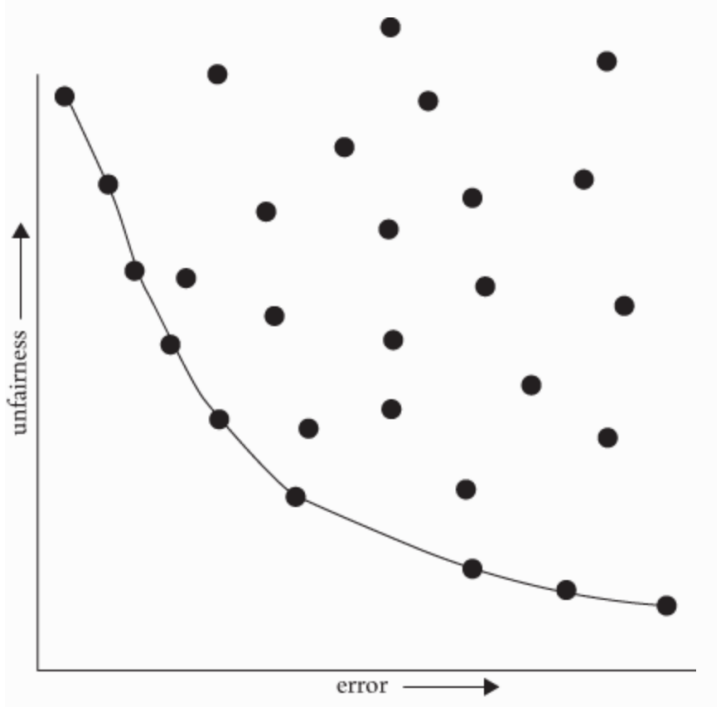
Brent Mittelstadt

This calls for a multipronged approach combining public policy, regulation, economic incentives, accountability as well as technical research.





> *Predictive accuracy and notions of fairness (and privacy, transparency, and many other social objectives) are simply different criteria, and that optimizing for one of them may force us to do worse than we could have on the other. This is a fact of life in machine learning.*



each point corresponds to a different model

Which point shall we choose?

This is a not a scientific question, but a question related to our ***values and social/political choices****.*

Technical Solutions and Tradeoffs

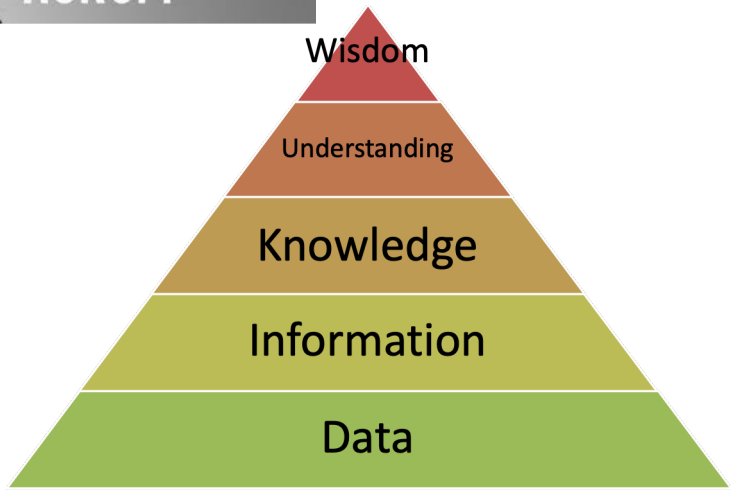
Using technology with wisdom

"We are now far too clever to be able to survive without wisdom."—E. F. Schumacher.

Can build it ≠ should build it

An ounce of information is worth a pound of data.
An ounce of knowledge is worth a pound of information.
An ounce of understanding is worth a pound of knowledge.
An ounce of wisdom is worth a pound of understanding.





Good for your self/tribe is not necessarily good for common good of humanity

Be careful of what you ask for







Values and Ends (Objectives)

Working Towards
Value Alignment



Thank you

jqadir@qu.edu.qa